Robotic Arm using Flexible Electrostatic Film Actuators

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***Abstract*—A 2-DOF robotic arm using flexible electrostatic film linear actuators named "*Dual Excitation Multiphase Electrostatic Drive*" (DEMED) has been developed. DEMED consists of centi-metric size stator and slider films, both are made of polyimide FPC (Flexible Printed Circuit) films. These films contain three-phase parallel electrodes with a width of 100μm at a regular interval of 200μm.**

DEMED has unique characteristics, such as lightweight, high-power, and flexible. Since DEMED consists of only flexible films, it can be bent easily at any direction and can be driven even if it is deformed. Besides, it can drive a load without any reduction gears.

The 2-DOF robotic arm has three actuators in its forearm and upper arm. In the upper arm, one actuator is arranged in bent state along the inside wall of the arm. In the forearm, a pair of actuators constructs antagonistic muscle-like structure. Since the structure has no reduction gears or driving belts, weight of the arm can be reduced.

The upper arm and forearm actuators generate thrust force of 23N and 10N at 1.6kV$_{0\text{-}p}$ respectively. Step responses of the two joints of the robotic arm were measured with feed back control system. Direct teaching playback control was performed utilizing the actuator's fine back-drivability. Finally, we discuss merits of utilizing DEMED.

***Index Terms*—robotic arm, lightweight actuator, flexible actuator, electrostatic actuator.**

I. Introduction

In late years, consumer robots such as humanoid robots or animal robots have been developed actively [1-4]. Those consumer robots required to be lightweight for safety. Since most of those robots employ electromagnetic actuators with reduction gears, their structures tend to be complicated and over-weight. The development of simple and lightweight robots demands more compact, high performance, and lightweight actuators. To meet the requirements of robotic actuator, several unique actuators have been developed[5-8]. Among those actuators, a electrostatic film actuator named *"Dual Excitation Multiphase Electrostatic Drive"* (DEMED)[8] has especially unique characteristics. Not only it is characterized by its high-power generation capability and lightweight structure, it is flexible so that it can be bent easily at any direction without losing its function. Besides, it can easily be controlled electrically.

Up to now, several investigations of DEMED have been conducted. It is revealed that DEMED has the capability of remarkable high-power per weight ratio up to 230W/kg and maximum speed of 1.4m/s[8]. A servo control strategy of the electrostatic actuator with a built-in position sensor was proposed and experimented[9]. Since DEMED can be bent and driven while it is deformed, thrust force characteristics of DEMED in bent state have been investigated through experimental measurements and theoretical analysis[10].

In this paper, we have fabricated 2-DOF robotic arm using DEMED. The robotic arm has two types of the electrostatic actuators. In the upper arm, one type of DEMED was arranged in bent state along the inside wall of the arm. In the forearm, a pair of DEMED constructs antagonistic muscle-like structure. The structure adopts no reduction gears or driving belts, it can reduce total weight of the robotic arm.

II. Dual Excitation Multiphase Electrostatic Drive (DEMED)

Fig. 1 shows basic components of DEMED. It consists of a pair of thin plastic films called slider and stator. Both films have three-phase parallel electrodes with a width of 100μm at a regular interval of 200μm. The base film and the electrode are made of polyimide and copper respectively. Slider and stator films are made using FPC (Flexible Printed Circuit) board technology.

Fig. 2 shows driving principle of DEMED. As shown in fig.2, when we apply three-phase ac voltage to the three-phase electrodes of slider and stator, traveling potential distributions are excited on their surface as displayed by bold lines. These two traveling potential distributions run toward opposite directions each other. Electrostatic force generated between two distributions thrusts slider film to make the slider run upon stator surface.

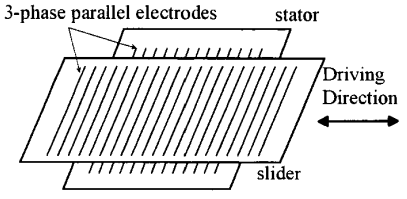


Fig. 1 Basic components of the electrostatic film actuator

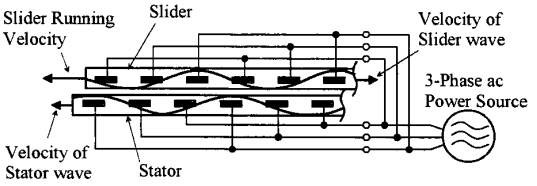


Fig. 2 Driving principle

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DEMED is a synchronous motor. Slider speed v increases in proportional to the frequency of applied ac voltage. The speed v is expressed as:

$$v=6pf \quad (1)$$

where p is electrode pitch, and f is frequency of there-phase ac voltage.

Fig. 3 shows a structure of DEMED. Since slider and stator films attract each other during driving, large frictional force is generated between films. To reduce the frictional force, a number of glass beads with a diameter of 20μm are sandwiched between films for lubrication. The gap between films is filled with dielectric fluid to prevent dielectric breakdown of the atmosphere. By using dielectric fluid, higher voltage can be applied to gain larger thrust force. Silicon oil (SH200, 50cSt, TORAY Dow corning Silicon Co.,Ltd) was used as dielectric fluid.

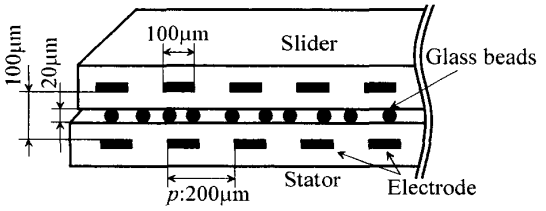


Fig. 3 Structure of the electrostatic film actuator

III. Robotic Arm Using Flexible Electrostatic Actuators

A 2-DOF robotic arm, which employs DEMED, has been fabricated. Fig. 4 shows a photograph of the robotic arm. The frame of the robotic arm is fabricated using cylinders of acrylic plastic. The total length of the robotic arm measures 500mm. The total weight is approximately 320g. Fig. 5 shows a schematic view of the robotic arm. On each rotating shaft of the joints, a potentiometer is equipped to detect the joint angle. Fig. 6 defines the joint angle of the robotic arm.

The robotic arm has two types of DEMED. In the upper arm, DEMED was arranged along the inside wall of the arm in bent state. In the forearm, a pair of DEMED constructs antagonistic muscle-like structure.

A. Fabrication of the upper arm actuator

Fig. 7 shows a photograph of stator and slider films that are utilized in the upper arm actuator. Slider and stator films measure 120mm wide by 120mm long by 200μm thick and 160mm long by 80mm wide by 200μm thick, respectively. The weights of stator and slider films are 3.7g and 3.4g, respectively.

Thrust force of DEMED can be enlarged by stacking films because thrust force increases in proportion to the contact surface area of slider electrode and stator one (referred as *effective electrode area*). For the upper arm actuator, three sets of stator and slider films are stacked to gain high-output. Effective electrode area of the actuator is approximately 150cm^2 in total.

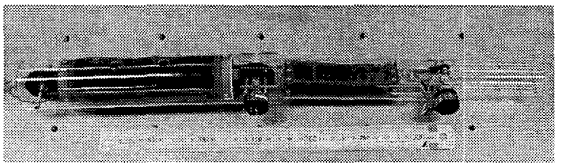


Fig. 4 Photograph of the robotic arm using electrostatic film actuators

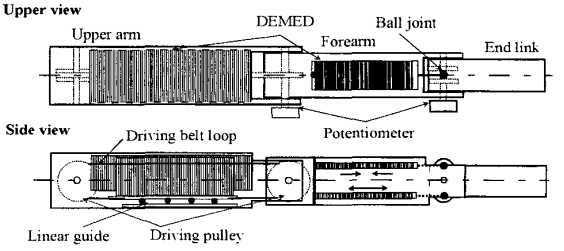


Fig. 5 Schematic view of the robotic arm

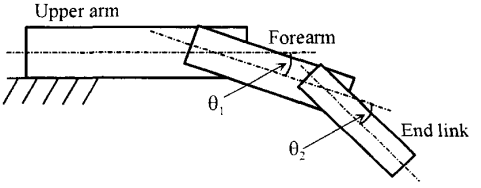


Fig. 6 Definition of the joint angle

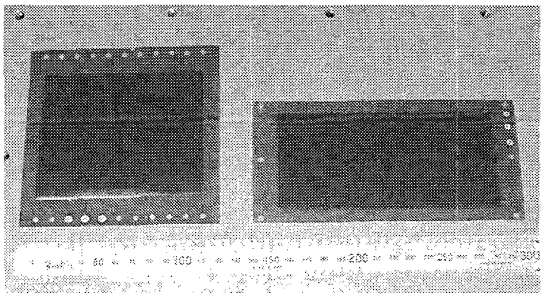


Fig. 7 Slider and stator films of the upper arm
Slider film (left) Stator film (right)

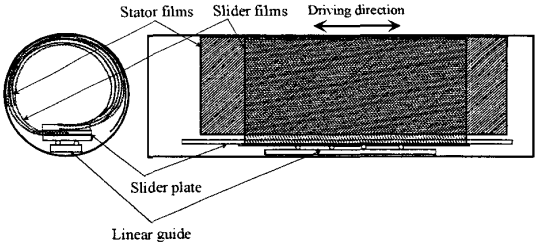


Fig. 8 Schematic view of the upper arm actuator

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Since the films are very thin and flexible, they can be easily rounded as shown in fig. 8. The upper arm actuator is rounded along inside wall of the arm to make utility space in the arm. Stator films are fixed to inside wall and slider films to a plate on a linear guide. By adopting this arrangement, large film can be set in a narrow space. Besides, we obtain wide free utility space inside the robotic arm. The utility space can be used to locate the wires and driving belt. As shown in fig. 9, the appearance of the robotic arm is very simple since wires and driving belts are located inside.

The actuator rotates the joint in both directions by using driving belt loop. Effective diameter of the driving pulley is 30mm. Range of rotation angle θ_1 is from -35 to +35 degrees. On the rotating shaft, a potentiometer is equipped to detect joint angle. The total mass of upper arm including all the parts measures 165g.

Differently from PM stepping motors or high-speed dc motors with reduction gears, DEMED has fine back-drivability. Since DEMED generates no holding force when applied voltage is zero, it can be moved easily by external force. When voltage is not applied to the actuator, frictional torque of the joint linking upper arm and forearm is only 2.1N·cm.

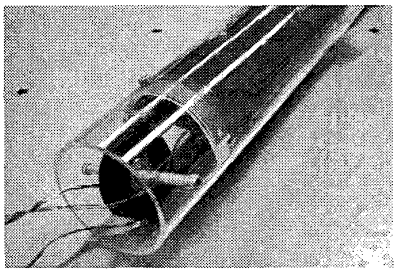


Fig. 9 Rear view of the upper arm

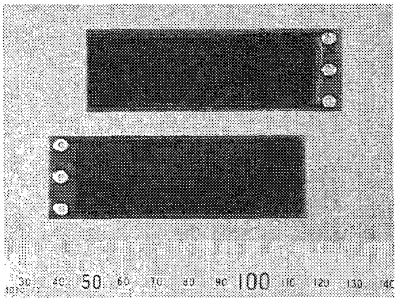


Fig. 10 Slider and stator films of the forearm

B. Fabrication of the forearm actuator

The forearm actuator should be lightweight because its mass becomes a load of the upper arm. In the forearm, an antagonistic muscle-like structure using DEMED was constructed. The structure is lightweight because it has no reduction gears or driving belts. The total mass of forearm including all the parts is 145g.

Stator and slider that are utilized in the forearm has the same structure. Fig. 10 shows a photograph of the films. The both films measure 29mm wide by 80mm long by 200μm thick and weigh 0.6g.

Fig. 11 shows the whole figure of the forearm actuator. The antagonistic muscle-like structure consists of two actuators. Each actuator consists of five sets of slider and stator films. Fig. 12 shows schematic view of the forearm actuator. Two actuators are driven using a common drive unit. A common three-phase voltage is applied to the actuators, with reversed connection to each other; when one of the actuator contracts, another one expands. The range of the joint angle is from -60 to +60 degrees. Corresponding to the joint angle, the length of the actuator changes form 107mm to 133mm. When the joint angle θ_2 is zero, length of both actuators become 120mm. The weight of the each actuator including dielectric fluid and mounting parts measures 20g.

The actuators are connected to the next link (end link) using ball joints. Joint torque is generated by contractile force of one of the actuators. This structure dose not involve backlash due to reduction gears. The forearm structure has fine back-drivability, as well as upper arm. Frictional torque

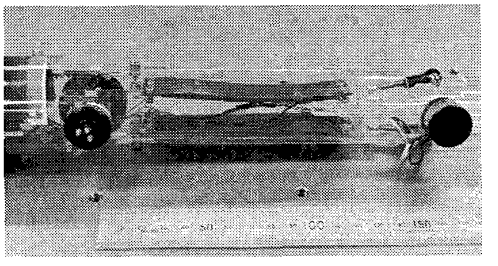


Fig. 11 Photograph of the forearm actuator

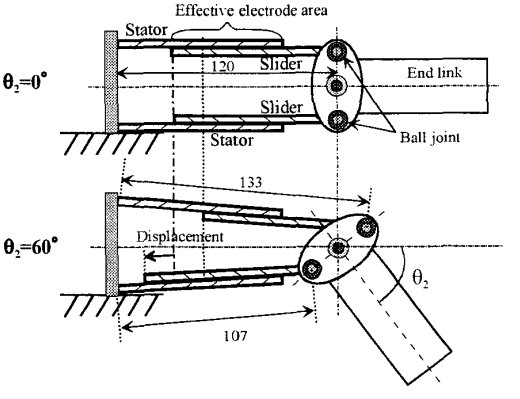


Fig. 12 Schematic view of the forearm actuator

of the joint between forearm and endlink is only 2.4N·cm, when a voltage is not applied.

IV. Experiments

A. Performance of the upper arm actuator

Thrust force was measured at a speed of 3mm/s and applied voltage ranging from $600V_{0\text{-}p}$ to $1600V_{0\text{-}p}$. We adopted

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a weight lifting experiments to measure thrust force. Thrust force was simply determined by finding maximum weight that the actuator was able to lift up. The weight lifting apparatus was built using pulley and wire.

The drive unit of the actuator consists of a function generator (AG1200, Yokogawa electric Corp.) as a source of the three-phase wave and three high voltage amplifiers (AS-123, NF Corp.).

The results of the measurements are shown in fig. 13. The data suggests that thrust force increases in proportion to the square of the amplitude of the applied voltage. The force weight ratio and joint torque are estimated to be 139N/kg and 34.5N·cm at $1.6kV_{0\text{-}p}$ respectively.

If we apply higher voltage, higher performance would be gained. However, the maximum performance was limited by electric breakdown between neighboring electrodes embedded in the films. Breakdown voltage of the films are approximately 2~3kV. For an application, which demands higher thrust force, we can enlarge thrust force by stacking more films.

B. Performance of the forearm actuator

We measured thrust force of the forearm actuator with respect to effective electrode areas, since effective electrode area changes as the actuator drives.

The results of the measurements are shown in fig. 14. Thrust force increases in proportion to effective electrode area.

When the joint angle θ_2 is zero, effective electrode area of both actuators become 40 cm^2, then, the actuator's force weight ratio and joint torque are estimated 48.3N/kg and 9N·cm at $1.6kV_{0\text{-}p}$ respectively.

C. Control of the robotic arm

A servo control system has been built to control the 2-DOF robotic arm. Although, a thrust force control method by which DEMED can be driven with no step-out was demonstrated[9], simpler control method was adopted in this study. In the control system, slider speed is controlled simply by changing frequency of ac voltage. Amplitude of voltage is kept constant.

The servo controller employs DSP system (DS1104, dSPACE,Inc.) and MATLAB Simulink system (The MathWorks,Inc.). Fig. 15 shows the block diagram of the feedback control system. Measured joint angle θ_b is compared with desired joint angle θ^* to calculate angle error, which is fed into PD controller. The PD controller output frequency of ac voltage. Based on the controller output, 3-phase oscillator generates three-phase ac voltage. The voltage is amplified by 1000 before being applied to the actuators. To prevent step-out of actuators, slider speed was limited to 50mm/s at the limiter, which is set between PD controller and 3-phase oscillator. The forearm and upper arm employed the same control system.

Fig. 16 and fig. 17 show the responses of the upper arm and the forearm against 10 degree step input, without any load. In this experiment, the robotic arm was placed so as to move in a horizontal plane. The amplitude of the ac voltage was kept at $1.0kV_{0\text{-}p}$.

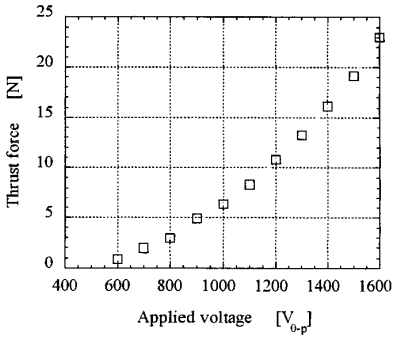


Fig. 13 Relationship between thrust force and applied ac voltage

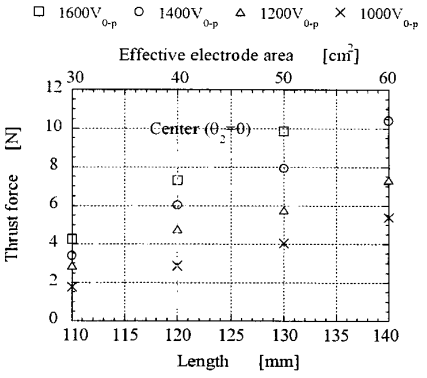


Fig. 14 Relationship of the effective electrode area and thrust force of the forearm actuator

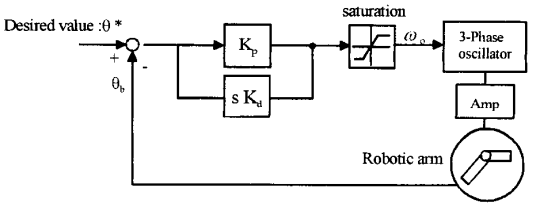


Fig. 15 Block diagram of the control system

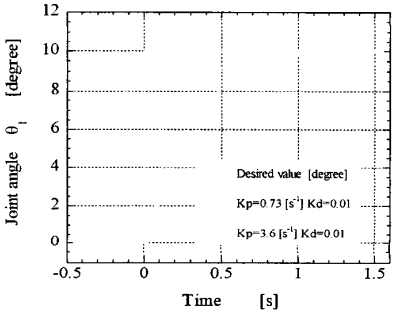


Fig. 16 Step responses of the upper arm

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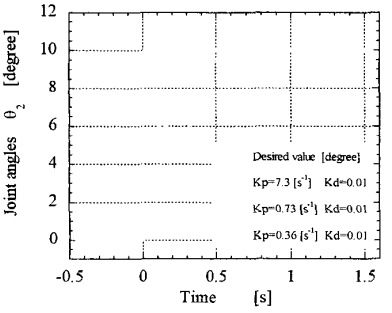


Fig. 17 Step responses of the forearm

Step responses of the forearm are relatively fine. The forearm can follow desired value of 10 degree in less than 0.5 second. On the other hand, a fluctuating behavior is observed in the data of the upper arm. This is probably due to deflection of the driving belt in the upper arm.

D. Teaching playback control of a pick and place task

When we make a robot do a certain work, we have to teach a task to the robot. There are several teaching methods, such as using teaching box, direct teaching, using CAD data file, and so on. Among these methods, direct teaching method is the easiest to perform for many applications.

We demonstrated direct teaching and playback control of a pick and place task. The sequence of postures is taught by moving end link by hand. Fig. 18 shows how we taught the task. Since the joints are moved smoothly, teaching work is very easy. The joint angles during teaching were measured by potentiometers and saved into a computer. The sampling interval of the data was 0.1 second.

Fig. 19 illustrates the sequence of the task. The sequence is as follows:

1. Pick up an object using a hook, which is attached to the tip of the end link.
2. Lift up the object to an upper shelf.
3. Put the object on the upper shelf (Unhook the object).
4. Return to an original position.

Fig. 20 shows a photograph of the robotic arm doing playback task. Fig. 21 shows the teaching data and playback data. Although, some delay is observed in the upper arm, the robot was able to perform the pick and place task. The delay is likely due to deflection of the driving belt or lack of stiffness of robotic arm structure.

V. Conclusions

A 2-DOF robotic arm using DEMED has been developed. In the upper arm, DEMED was arranged in the bent state to make utility space in the robotic arm. In the forearm, a pair of DEMED constructs antagonistic muscle-like structure. The structure has no reduction gears or driving belts and can reduce weight of the robotic arm. The upper arm actuator and forearm actuator generate joint torque of 34.5N·cm and 9N·cm at $1.6kV_{0\text{-}p}$ respectively.

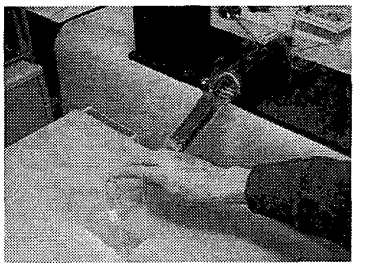


Fig. 18 Photograph of the task teaching

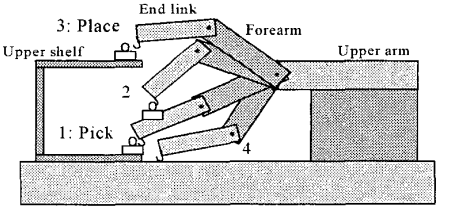


Fig. 19 The sequence of the pick and place task

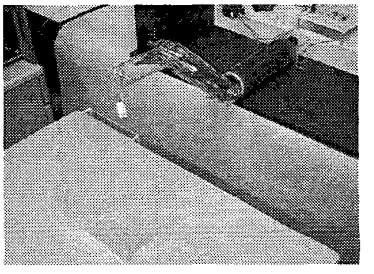


Fig. 20 Photograph of the playback task

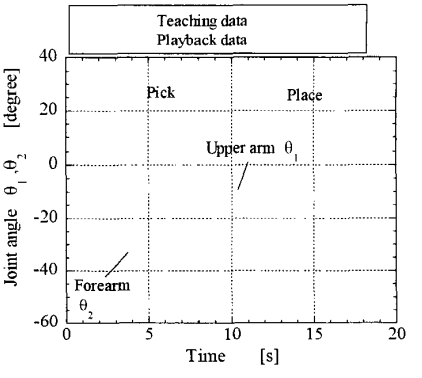


Fig. 21 Teaching data and playback data

A servo-control system was built to measure the performance of the robotic arm. Step response of the forearm actuator is superior to that of the upper arm actuator. It is likely that the upper arm actuator employed driving belt, whereas forearm actuator employed direct driving. Finally,

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direct teaching playback control was performed utilizing DEMED's fine back-drivability.

In the following, we sum up merits of utilizing DEMED for consumer robot.

1. DEMED can work even if it is deformed: DEMED can be set in various spaces.
2. DEMED is very lightweight: DEMED can be arranged to the serial link robots easily.
3. DEMED requires no reduction gears: we can control robots precisely by direct driving.
4. DEMED (slider) can be moved freely by external force when voltage is not applied: DEMED has fine back-drivability.
5. DEMED is not damaged even if strong external force is applied because it has no gears.

Some consumer robots require the fifth item because they will receive unexpected strong external force while human uses it. If an actuator has high reduction gear ratio, reduction gears would be damaged easily when the actuator is exposed to unexpected strong external force.

As the future direction of this study, we will consider better effective arranging methods of the actuators. Also, we are planning to develop end effecters utilizing DEMED.

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